AgentSynth: Scalable Task Generation for Generalist Computer-Use Agents

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Paper version: arXiv 2506.14205v2 · CUA Radar Reel poster, not made or endorsed by the authors

Problem

> *Current LLM agents struggle to generate reliable trajectories for complex tasks.*

Human curation does not scale: OSWorld took ~1,800 person-hours for 412 tasks; prior synthesis pipelines stay mostly web-only.

Why synthesis is hard

1. **Hard to generate directly.** One-shot instructions: 11% get a trajectory.
2. **Diversity.** Persona-seeded, chained subtasks: 6,000+ tasks.
3. **A difficulty dial.** Summarize the first n subtasks → level n.

Evaluation setup

OSWorld Ubuntu VM · 1920×1080 screenshots · GPT-4.1 proposer · CU-preview grounding · 10-step subtask cap · $0.60 per trajectory

- **Eval:** 50 tasks per level; bare LLMs write pyautogui; S3 / S2.5 scaffolds; humans: 20 tasks.
- **Online in OSWorld**, judged by the paper's LLM verifier; InSTA port: 194 tasks.

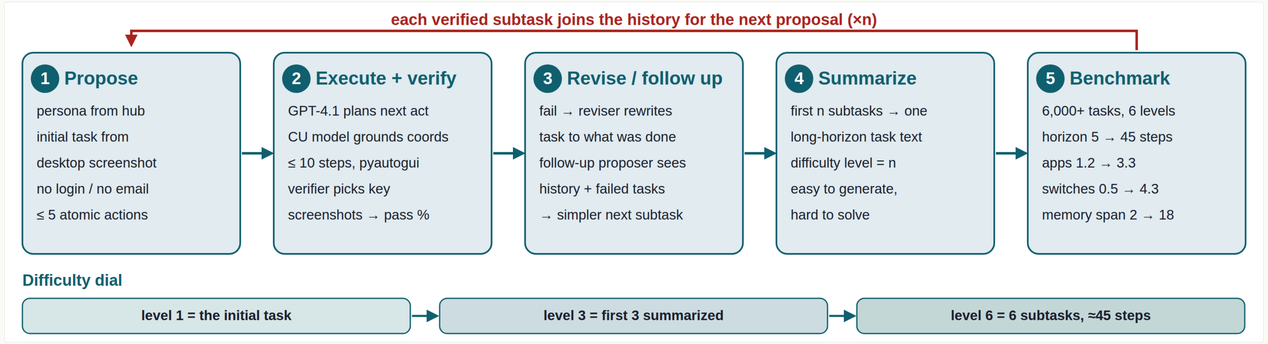
Our read (CUA Radar)

64	★★	online
score of 100	band	evidence level

- **Reports:** online OSWorld eval; cost; ablations; verifier stress test; code.
- **Flags:** held-out apps unclear; no seeds / variance; own LLM verifier judges success.
- Pattern K15. Authors' numbers, not rerun.

Method: chain verified subtasks, summarize into level-n tasks

- **Sees / acts:** full-screen 1920×1080 screenshot in; GPT-4.1 plans, computer-use-preview grounds clicks and keys via pyautogui.
- **Six LLM roles:** proposer (persona-conditioned), executor, verifier, reviser, follow-up proposer, summarizer.
- **Platform:** OSWorld Ubuntu VM desktop; an InSTA web port reuses the same pipeline.
- **Signal:** binary + % completion from the verifier; failed tasks steer the next proposal simpler.



Our redraw (not the paper's figure): the generation loop, the failure-feedback edge, and the difficulty dial.

Results: success rate (%) by difficulty level

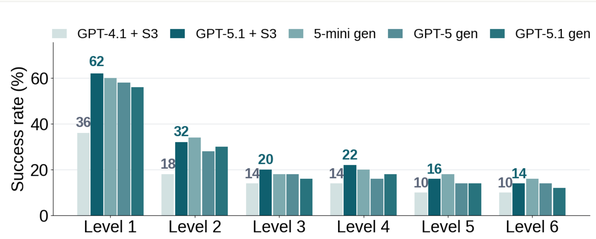
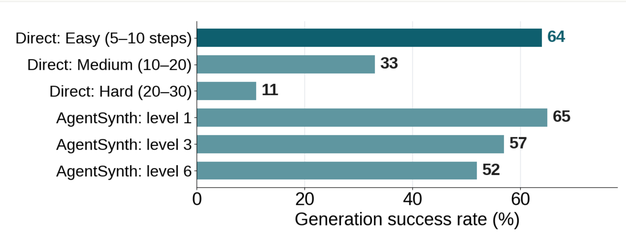


Table 6, redrawn, verifier-judged online in OSWorld: first two series use the GPT-4.1-generated dataset; 'gen' rows swap the synthesis model (eval: GPT-5.1 + S3).

o4-mini: 18% at level 1 → 4% at level 6; humans 70% on level 6

SO WHAT → *All settings decline with level; no seeds or CIs reported. Bare GPT-4.1 scores zero beyond level 3.*

Ablation: information asymmetry (Table 5)



Redrawn from Table 5: generation success = a ground-truth trajectory exists; evaluation by Agent S3 with GPT-5.1; direct baselines target 5–10 / 10–20 / 20–30 actions.

SO WHAT → *Direct generation collapses on hard tasks (11%); chaining holds 52% — while eval success falls 62% → 14%. Retained direct tasks stay easy for the eval agent (60 / 51 / 48), so filtering alone cannot yield a hard benchmark; stable generation at every level is what makes 6,000+ tasks cheap to collect.*

Headline Numbers

18% → 4%

o4-mini, level 1 → 6

humans: 70% on level 6

70% human success, level 6

$0.60 cost per trajectory

6,000+ tasks synthesized

The paper's own controls

CONTROL	VALUE	CONTEXT
Verifier stress test	12% / 94%	near-miss / benign
Human audit (100 tasks)	91/90/94/88%	feas./coher./pers./ver.
Generator swap (Table 6)	62 → 56 at L1	GPT-4.1→GPT-5.1 gen

Success is judged by the authors' LLM verifier (88% human agreement).

Takeaway

Chain verified subtasks, then summarize: tasks cheap to build, hard to solve; every tested model declines with level.

gen 65→52% (L1→L6) · eval 62→14% (L1→L6) · >60% use 2+ apps · L6: 40–60 steps

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